

SOYAL

Communication Protocol

Item No.	AR-721H	Version 3.X
	AR-721HV3	Version 5.X
	AR-727HV3	Version 6.X
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1 Communication Protocol

All SOYAL® devices use the same protocol, only the commands given vary depending on different devices

1.1 Preparing Connect:

Prior to connection to AR-721H, the node number of the device has to be known. The node number could be checked from the keypad as shown on the table below:

Steps	AR-721H	AR-727H
Step 1:(Enter into EDIT MODE)	* 1 2 3 4 5 6 #	* 1 2 3 4 5 6 #
Step 2:(Setting Node ID)	0 0 * 0 0 1 #	3 → 1 → 1 #
Step 3:(Exit)	* #	* * * #

1.2 Data format:

Baud Rate			9600 · N · 8 · 1			
Data Format			Binary HEX Data			
Data Packet						
Head	Length	Destination ID	Command Code	Data ...	XOR	SUM
8bits	8bits	8bits	8bits		8bits	8bits

1.3 Data package:

	Size (Byte)	Description	XOR 0xFF	SUM 0x00
Head	1	0x7E	No	No
Length	1	Data Length Indicator which denotes the length from Destination to the end including XOR and SUM	No	No
Destination ID	1	Destination Node ID 00: Reserved for the bus master Send data to PC FF: Broadcast to each reader	Yes	Yes
Command Code	1	Instruction Command	Yes	Yes
Data	00 ~ D0	Length of Data Block which varies depending on instructions	Yes	Yes
XOR	1	To XOR each byte from Destination ID to Data with 0xFF		Yes
SUM	1	TO sum each byte from Destination to XOR with 0x00. If the summary is greater than 0xFF, it should keep the low byte.		

Example : Polling Status from Node. 1

Head	Length	Destination	Command	Data ...	XOR	SUM
0x7E	0x04	0x01	0x18		0xE6	0xFF

$$\text{XOR} = 0xFF \wedge 0x01 \wedge 0x18 = 0xE6$$

$$\text{SUM} = 0x01 + 0x18 + 0xE6 = 0xFF$$

Node ID needs to set from the keypad . Please refer to the command list for- 00*

1.4 Echo Code :

Command	Description	Note
03h	Echo requested data	7E xx 00 03 [Data...] XOR SUM
04h	Echo command acknowledged (ACK)	7E 04 00 04 FB FF
05h	Echo command unacknowledged (NACK)	7E 04 00 05 FA FF
06h	Echo authentication failed (AUTHERR)	7E 04 00 06 F9 FF
07h	Echo no tags presented (NOTAG)	7E 04 00 07 F8 FF
08h	Echo not login (NOT LOGIN LEVEL2)	7E 04 00 08 F7 FF
09h	Echo CRC8 check error for ReadBlockCRC8	7E 04 00 09 XX XX
0Ah	Echo not authenticated	
0Bh	Echo authentication layer rejected	

※ AR727H Data appended of ACK/NACK:

Description	The ACK/NACK echo of AR727H will followed device information after (0x04/0x05) like that: Echo Package: 7E Len 00 04/05 Node Type Sta0 Sta1 Option0 Version XOR SUM
Node	Controller Node ID of AR727H
Type	Controller Type of AR727H, 0x23(125KHz), 0x24(13.56MHz)
Sta0	Bit7 : Keypad Lock/Unlock (1/0) Bit6 : Door Relay On/Off (1/0) Bit5 : Alarm Relay On/Off (1/0) Bit4 : Arming Active/Inactive (1/0) Bit3 : Alarm Active/Inactive (1/0) Bit2 : Reserved Bit1 : Exit button Close/Open status (1/0) Bit0 : Door Sensor Close/Open status (1/0)
Sta1	Bit7 : Forced Open Alarm

2 Command List

2.1 Operation Mode

There are two operation modes one this device.

Networking Mode:

The device will detect whether received Command 18H from Host before time out. If host sent 18H and device received the message, the device will pass tag ID to host and wait for response from host echo ACK , NACK or release commands. If host did not response code to device within 25 seconds, the device will display “network communication error!” with 7 beeps. If the device did not receive command 18H within 10 seconds, the device will automatically switch to stand-alone mode.

Release command: 04H, 05H, 84H.

Stand-alone Mode:

All access events on the device will be identified itself. The host can get transaction log via command 25H. In stand-alone mode, user can select operation mode between M4, M8 and M6. But in networking mode the user only can select M4 or M8.

2.1.1 18H Get Device Status (Event Polling)

Send	Value	Description
Head	7E	Leading Code
Length	04~17	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Destination Node ID, it refers node 1 here
Command	18	Get the current device status (Event Polling)
	*	[Second], Optional data
	*	[Minute]
	*	[Hour]
	*	[Weekday]
	*	[Day]
	*	[Month]
	*	0x00
	*	[Weekday]
	*	[Year]
	*	0x00
XOR	E6	$\text{XOR} = \text{FF} \wedge 01 \wedge 18 = \text{E6}$
SUM	FF	$\text{SUM} = 01 + 18 + \text{E6} = \text{FF}$

P.S.: The optional data can support real time clock synchronization between slave and host.

Echo	Value	The device respond depending on different status as follows (1) Standby status w/o event log (2) User key buffer status for PIN only or card and PIN (3) Card flashing Status (4) Standby status with event log
Head	7E	Leading Code
Length	??	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Destination	00	Destination Node ID. 00H mean send to host (PC or AR716E)
Function Code	09	09H is echo reader status code.
Source	01	Source Node ID. (Who send);
Event	Event	Event type of followed data (please refer to 2.1.2)
Data Field	Data 0	Data 0
	Data 1	Data 1
	Data 2	Data 2
	Data 3	Data 3
XOR		Check byte
SUM		Check byte

2.1.2 Device Status Define

Event	Meaning	Description of Data Bit																																				
00	AR721H Status	Neither card nor key event, just echo device I/O status. If event log buffer is not empty. The oldest event will be appended and followed the Data3. *) If event log appended, the package length will more then 36(24h) bytes. Check the package length to identify if event log appended or not. *) The appended event will be fixed in 26 bytes format and start in event code. Check “Message File Strcuture.doc” to get field definition.																																				
00	AR727H Status	*Same with AR721H																																				
81	AR721W Status	*Same with AR721H																																				
82	AR721D Status	*Same with AR721H																																				
83	AR721Q Status	*Same with AR721H																																				
01	4/5 Keys pressed Note : Mode4: 5 keys pressed, Mode8: 4 keys pressed	Data 0 : The 5th Key data. Set Bit7 while working on Mode 8. Data 1 : Input value’s MSB Data 2 : Input value’s LSB Data 3 : Undefined default value=0 Data 4 : Device parameters. (Setting by command: 20*XXX#) Data 5 : 401RO16’s parameter (24*xxx#) Data 6 : Undefined default value=0 Controllers are allow to identify the value Data 7~ Data 11 : M4 : The Key data M8 : Data 7~Data 10 = Key data , Data 11 = 00 Data 12 : Undefined default value=0																																				
02	New Card Present UID4: Tag ID Bits(39~32) UID3: Tag ID Bits(31~24) UID2: Tag ID Bits(23~16) UID1: Tag ID Bits(15~08) UID0: Tag ID Bits(07~00)	The inner code of the card is 40bits in length which can be tabled 7E xx 00 09 01 02 Dat0 UID3 UID2 Dat3 Dat4 UID1 UID0 UID4..... XOR SUM below: <table border="1"><tr><td>MSB</td><td colspan="4">Inner Code</td><td>LSB</td></tr><tr><td>ID</td><td colspan="2">Site Code</td><td colspan="2">Card Code</td><td></td></tr><tr><td></td><td>High</td><td>Low</td><td>High</td><td>Low</td><td></td></tr><tr><td>ID</td><td>Site Hi</td><td>Site Lo</td><td>Card Hi</td><td>Card Lo</td><td></td></tr><tr><td>8 Bits</td><td>8 Bits</td><td>8 Bits</td><td>8 Bits</td><td>8 Bits</td><td></td></tr><tr><td>39~32</td><td>31~24</td><td>23~16</td><td>15~08</td><td>07~00</td><td></td></tr></table> Data 0 : Time & Attendance , Bit7~Bit5 : Time & Attendance Selection Other LSBs refer to “Exit Input” Data 1 : High byte of site code (bit 15~08)	MSB	Inner Code				LSB	ID	Site Code		Card Code				High	Low	High	Low		ID	Site Hi	Site Lo	Card Hi	Card Lo		8 Bits	8 Bits	8 Bits	8 Bits	8 Bits		39~32	31~24	23~16	15~08	07~00	
MSB	Inner Code				LSB																																	
ID	Site Code		Card Code																																			
	High	Low	High	Low																																		
ID	Site Hi	Site Lo	Card Hi	Card Lo																																		
8 Bits	8 Bits	8 Bits	8 Bits	8 Bits																																		
39~32	31~24	23~16	15~08	07~00																																		

		Data 2 : Low byte of site code (bit 07~00) Data 3/4 : The MSB / LSBs of the value input before card flashing, it would show the previous value if there is not value input before flashing Data 5 : High byte of card code (bit 15~08) Data 6 : Low byte of card code (bit 07~00) Data 7 : ID Code(Bits(39~32) of EM4001 Chi Data 8 : Bit Selection (20*xxx#) Data 9 : Bit7:1/0 From WG/Controller Bit6: 1/0 Setting Forced Open Alarm
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Echo Status Field for AR721H , AR727H		Code : 00H
Data 0	Bit7 : Keypad Lock/Unlock (1/0) Bit6 : Door Relay On/Off (1/0) Bit5 : Alarm Relay On/Off (1/0) Bit4 : Arming Active/Inactive (1/0) Bit3 : Alarm Active/Inactive (1/0) Bit2 : Reserved Bit1 : Exit button Close/Open status (1/0) Bit0 : Door Sensor Close/Open status (1/0)	
Data 1	Bit7 : Forced Open Alarm Bit6 : Bit5 : Set to 1 for Editing Status.	
Data 2	Device parameters (Command 20 * xxx#)	
Data 3	Bit 7: Set to 1 for Card Present. (Tag still in the RF Filed) Bit 6: Set to 1 if in the Free Open zone	
Data 4 ~ 29	If there has any transaction in the queue buffer, the device will append the last event follow the Data 3, and then the host can use command 37 to remove this log.	

Echo Status Field for AR721W		Code : 81H
Data 0	Bit 7 : Relay On/Off (1/0) Bit 6 : Card Present. (Tag still in the RF Filed) Bit 5 : DO1 On/Off status (1/0) Bit 4 : DO0 On/Off status (1/0) Bit 3 : DI3 On/Off status (1/0) Bit 2 : DI2 On/Off status (1/0) Bit 1 : DI1 On/Off status (1/0) Bit 0 : DI0 On/Off status (1/0)	
Data 1	Bit7 : Forced Open Alarm	
Data 2	Device parameters (Command 20 * xxx#)	
Data 3	Tag Type (Tag still in the RF Filed) 0: No tag 1: Ultra-light 2:S50 3:S70 4:15693 5: DESFire	

Data 4 ~ 7	If Tag present this field will be 32bits UID and event data followed from data 8 or 11 otherwise will be null and event data followed from data 4	
Data 8 ~ 10	If ultra light present this will be ultra light Tag UID (Bit 55 ~ 32) and event data followed from data 11	
Data 4 ~ 31 (Data 8~35) (Data 11~38)	If there has any transaction in the queue buffer, the device will append the last event follow the Data 3, and then the host can use command 37 to remove this log.	

(1) Standby Status of the Device

Example : Suppose arming is active, the information below is what is given from the device when "Request to Exit" button is pressed and the door relay is activated

Echo	Value	Description																
Head	7E	Leading Code																
Length	0A	Data Length Indicator which denotes the length from Node to the end including XOR and SUM																
Node	00	The value 00 is fixed, the message would be sent to PC from the device																
Function	09	The value 09 is fixed, and represents the enquiry command sent to the controller from the device																
Reader ID	01	Original Node ID to send message																
Event	00	The device is under standby status																
Data Field	52	Data0: can be converted into 8bits as below <table border="1"><tr><td>Bit7</td><td>Bit6</td><td>Bit5</td><td>Bit4</td><td>Bit3</td><td>Bit2</td><td>Bit1</td><td>Bit0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> Device Status: Bit 6 : Activating Door Relay Bit 4 : Arming Active Bit 1 : Pressing "Request to Exit" button	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	0	1	0	1	0	0	1	0
		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0									
		0	1	0	1	0	0	1	0									
		00	Data1: Setting Forced Open Alarm															
		18	Data2: (20 * xxx#)															
00	Data3: no tag in the RF Field																	
XOR	BD	XOR=FF^00^09^01^00^52^00^18^00=BD																
SUM	31	SUM=(00+09+01+00+52+00+18++00+BD) · FF=31(Select the LSB)																

ST0	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
AR721H	Keyboard	Door	Alarm	Arming	Alarmed		Egress	Door
AR721H	Locked	Relay	Relay				Button	Opened
AR747H								
AR721D	Battery OK	Lock:0 Unlock:1		Internal Locked				
AR721W	Relay On	Card Present	DO1	DO0	DI3	DI2	DI1	DI0
QFM02	Relay On	Card Present	DO-1	DO-0		Door Open	Egress	DI0

(2)Keypad Status of the Device

Example : The information below is what is given from the device when keying in 12345			
Echo	Value	Description	
Head	7E	Leading Code	
Length	0D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to host	
Function	09	The value 09 is fixed, and represents the enquiry command sent to the controller from the device	
Reader ID	01	Original Node ID to send message	
Event	01	Control code for keypad status	
Data Field	05	The 5th Key data	In Mode8 the bit7 of Data0 will be set. When keying in “1 2 3 4 #” from the keypad, this will be converted to 0x04D2, 0x04 will be put in Data1, 0xD2 will be put in Data2, and Data0 will be 0x80.
	30	Data 1 (0x04)MSB	
	39	Data 2 (0xD2)LSB	
	00	Data 3 fixed to 00	
	18	Data 4 (20*xxx#)	
	14	Data 5 (24*xxx#)	
	00	Data 6 fixed to 00	
		Data 7~Data11	
		Data 12	
XOR	F6	XOR=FF^00^09^01^01^05^30^39^00^00^18^14^00 =F6	
SUM	9B	SUM=(00+09+01+01+05+30+39+00+00+18+14+00+F6) · FF=9B(Select the LSB)	

(3)Card Present Status of the Device

Example : The information below is what is given from the device when flashing a card			
Echo	Value	Description	
Head	7E	Leading Code	
Length	13	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Function	09	The value 09 is fixed, and represents the enquiry command sent to the controller from the device	
Reader ID	01	Original Node ID to send message	
Event	02	Card Flashing Status	
Data Field	1E	Duty Code	Bit 7~5 of Data0 is duty code.
	63	UID(Bit31~24)	Each EM tag has 40 bits UID code.
	6B	UID(Bit23~16)	The 1 st byte is ID Code (Bit 39~32)
	04	Value Hi	The next four bytes are site code (Bit 31~16) and card code(bit15~00). In Mifare mode the ID code always is 0.
	D2	Value Lo	
	B8	UID(Bit15~08)	Device parameter:
	B4	UID(Bit07~00)	Bit7: Enable Anti-pass-back (1/0)
	04	ID Code	Bit6: Input/Output door (1/0)
	10	Device Parameters	Bit5: Master/Slave device (1/0)
	00	User Status	Bit4: Enable egress button (1/0)
Used after Mifare 6.X	8A	Identify	8AH for Mifare Tag
	22	Data11 TagType	Media(Bit 7~5) and Tag(Bit4~0) type. Media Type: 1:LAM, 2:UIM, 3:CIM, 4:SIM. Tag Type: 1: Ultra Light 2:S50 3:S70 4:15693 5: Desfire

	50	Data12 Flag	For SOR function parameter (Flag), Please refer to Mifare Key Device Layer. Bit 7 : will enable Date Time Check Bit 6 : will enable SOR layer Bit 5 : Reserved Bit 4 : will enable Decrement value check Bit 3 : Reserved Bit 2 : will enable Global Media Check Bit 1 : Reserved Bit 0 : will enable Lift data check
	4 bytes	Data 13~16	If Ultra Light presented this filed will be Tag UID(Bit 63~32)
XOR			
SUM			

Ultra Light present: Type 01, UID is 00.00.10.D9.C2.91.04

7E 17 00 09 01 02 09 [D9 C2] C0 05 [91 04] 00 18 00 8A [01] 00 [00 00 00 10] 34 F1

S50 Present: Type 02, UID is 6B.2F.87.66

7E 13 00 09 01 02 09 [6B 2F] C0 05 [87 66] 00 18 00 8A [02] 00 0C 11

2.2 04H Prompt Accepted Message

Echo	Example Value - AR721	Example Value - AR727	Description	
Head	7E	7E	Leading Code	
Length	04	09	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	01	Node ID of destination, the node id is 01 here	
Command Code	04	04	Informing the device to send correct message (1) A sound from the beeper (2) A flash from the green LED (3) Door opening operated by relay (4) "UID-H + UID-L" showed on LCD (AR-727H only)	
SS		A0	SS: Auxiliary Command(AR-727H only) 01: "Force On" showed on LCD, relay doesn't activate 02: "OK" showed on LCD, relay activates 03:"Force Off" showed on LCD, relay doesn't activate 13:Attaching information about lift stops available for the user A0:"UID-H + UID-L" showed on LCD	
XX		00	If the card presented is valid, please insert UID(bit15~8), others can be 0 °	
XX		00	If the card presented is valid, please insert UID(bit7~0), others can be 0 °	
UID-H		01	MSB of UID (AR-727H only)	"UID-H + UID-L" showed on LCD (00300 OK) 300 converts to 0x012C 0x01 will be placed in UID-H 0x2Cwill be placed in UID-L
UID-L		2C	LSB of UID (AR-727H only)	
Lift LL			Lift stops available for the user (Bit 07~00)	
Lift LH			Lift stops available for the user (Bit 08~15)	
Lift HL			Lift stops available for the user (Bit 16~23)	
Lift HH			Lift stops available for the user (Bit 24~31)	
XOR	FA	77	XOR=FF^01^04^A0^00^00^01^2C =77	
SUM	FF	49	SUM=(01+04+A0+00+00+01+2C+77) · FF=49(LSB)	

Note1: This command belongs to unresponsive commands which wouldn't respond either correct or wrong information.

Note2: The information about lift can be ignored if there is no command related to stops.

2.3 05H Prompt Invalid Message

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	05	Informing the device to send wrong message (1) Two sounds from the beeper (2) Two flashes from the red LED
XOR	FB	$XOR = FF \oplus 01 \oplus 05 = FB$
SUM	01	$SUM = (01 + 05 + FB) \cdot FF = 01 (LSB)$

Note: This command belongs to unresponsive commands which wouldn't respond either correct or wrong information.

2.4 09H Prompt Keying-in Password

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	09	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	09	Informing the device to send message about keying in password (1) Four sounds from the beeper Behind the command code, Data 0 should be 0x40, Data 1&2 are undefined and Data 3&4 are the number sent from the device. The number will be converted to a value with 5 digits (decimal) and showed on LCD (AR-727H only).	
Data 0	40	0x40(fixed)	
Data 1	00	Undefined, value=00 is fixed	
Data 2	00	Undefined, value=00 is fixed	
Data 3	00	Num H MSB	"Num H Num L" showed on LCD (00088 OK)
Data 4	58	Num L LSB	
XOR	EF	XOR=FF^01^09^40^00^00^00^58 =EF	
SUM	91	SUM=(01+09+40+00+00+00+58+EF) · FF=91(LSB)	

Invalid Tag examples

Host Polling :7E 0D 01 18 20 2B 0B 08 04 00 01 12 00 F9 87

Slave Echo :7E 10 00 09 01 02 0B 10 01 00 00 10 01 77 C8 00 41 B9

Host Response:7E 0B 01 05 00 10 01 3A 98 10 01 59 53

Card Only Tag example:

Host Polling :7E 0D 01 18 0D 2C 0B 08 04 00 01 12 00 D3 4F

Slave Echo :7E 10 00 09 01 02 0B 00 65 00 00 0F C5 42 C8 00 DB 35

Host Response:7E 0D 01 04 00 0F C5 00 4E 00 00 00 65 1B A7

Card and PIN Tag example:

Host Polling :7E 0D 01 18 17 3A 0B 08 04 00 01 12 00 DF 73

Slave Echo :7E 10 00 09 01 02 0B 04 D5 00 00 B8 26 01 C8 00 78 0F

Host Response:7E 0D 01 09 40 B8 26 00 59 16 2E 04 D5 99 37 ;This tag is Card & PIN

Host polling :7E 0D 02 18 17 3A 0B 08 04 00 01 12 00 DC 71 (Polling next Node ...)

....

Host polling :7E 0D 01 18 1B 3A 0B 08 04 00 01 12 00 D3 6B (The Node who got Tag)

Slave Echo :7E 18 00 09 01 03 0B 00 59 02 C8 16 2E 05 06 07 08 0B 00 E7 A7 8D 8C 7F 6D 37 (User Input)

Host Response:7E 0D 01 04 08 B8 26 00 59 00 00 04 D5 E4 01

Host polling :7E 0D 02 18 1B 3A 0B 08 04 00 01 12 00 D0 69 (Polling next Device)

2.5 20H Write EEPROM

Echo	Example Value	Description
Head	7E	Leading Code
Length	0F	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	20	Write EEPROM can up to 8bytes at a time (721HV3/727HV3 can be up to 32 bytes at a time) Please refer to section 3.5 for more details
AddrH	00	Writing address in EEPROM
AddrL	80	
Bytes	08	8 bytes
Data 0	11	Byte 1
Data 1	22	Byte 2
Data 2	33	Byte 3
Data 3	44	Byte 4
Data 4	55	Byte 5
Data 5	66	Byte 6
Data 6	77	Byte 7
Data 7	88	Byte 8
XOR	DE	$XOR = FF^{01^{20^{00^{80^{08^{11^{22^{33^{44^{55^{66^{77^{88}}}}}}}}}} = DE$
SUM	EB	$SUM = (01 + 20 + 00 + 80 + 08 + 11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + DE) \cdot FF = EB(LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function Cod	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF^{00^{04^{01}} = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.6 12H Read EEPROM

Echo	Example Value	Description
Head	7E	Leading Code
Length	07	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	12	Read EEPROM up to 8bytes at a time (721HV3/727HV3 can be up to 32 bytes at a time) Please refer to section 3.5 for more details
AddrH	00	Reading address out from EEPROM
AddrL	80	
Bytes	08	8bytes
XOR	64	$XOR = FF \oplus 01 \oplus 12 \oplus 00 \oplus 80 \oplus 08 = 64$
SUM	FF	$SUM = 01 + 12 + 00 + 80 + 08 + 64 = FF$

Echo	Value	Description
Head	7E	Leading Code
Length	0D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	02	Message sent from the device
Reader ID	01	Reader ID
Data Field	11	Data 0
	22	Data 1
	33	Data 2
	44	Data 3
	55	Data 4
	66	Data 5
	77	Data 6
	88	Data 7
XOR	74	$XOR = FF \oplus 00 \oplus 02 \oplus 01 \oplus 11 \oplus 22 \oplus 33 \oplus 44 \oplus 55 \oplus 66 \oplus 77 \oplus 88 = 74$
SUM	DB	$SUM = (00 + 02 + 01 + 11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + 74) \cdot FF = DB(LSB)$

2.7 21H Relay Control

Echo	Example Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	21	Relay control (on or off), no indication for it
Data	82	Activating Relay (refer to 2.7.1)
XOR	5D	$XOR = FF \oplus 01 \oplus 21 \oplus 82 = 5D$
SUM	01	$SUM = (01 + 21 + 82 + 5D) \cdot FF = 01(LSB)$

2.7.1 Relay Control Parameters

Data 0	Description	Note
0x00	Checking I/O Status	
0x01	Data1 =(1/0)keypad lock/unlock	Only suits 6V2 or above
0x02	Data1/2 =Interval between card flashing (about 10ms)	Only suits 6V2 or above
0x80	Activating Arming. (Enter armed state)	
0x81	Disable armed state	
0x82	Door Relay On	
0x83	Door Relay Off	
0x84	Activating Door Relay only for a certain time	* 1
0x85	Alarm Relay On	
0x86	Alarm Relay Off	
0x87	Activating Alarm Relay only for a certain time	* 1
0x88	On 721W Digital Output. 7E LEN DID 21 88 BIT	BIT2~0 : DO2~0
0x89	Off 721W Digital Output. 7E LEN DID 21 88 BIT	BIT2~0 : DO2~0

Note1:Activating the relay for a certain time and then inactivating it. The relay time can be set from the keypad. Please refer to the operation manual for the command 02* and 03*.

Ex: Get I/O status

7E 05 DID 21 00 XOR SUM

Set DO-0 on at 721W

7E 07 DID 21 88 STATE XOR SUM ; STATE is bit mapping to 721W DO. If state equal to 0x03, the DO0 & DO1 will set ON

Off DO-0 on at 721W

7E 07 DID 21 89 STATE XOR SUM ; STATE is bit mapping to 721W DO. If state equal to 0x02, the DO1 will set OFF

Set DO-0 on at 721QFM

7E 07 DID 21 88 STATE XOR SUM

STATE is bit mapping to 721QFM DO.

Off DO-0 on at 721QFM

7E 07 DID 21 89 STATE XOR SUM;

STATE is bit mapping to 721QFM DO.

Bit0: DO0

Bit1: LED Green

Bit2: LED Red

BIT3: DO1

Echo	Value	Description	
Head	7E	Leading Code	
Length	09	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Function	03	Response to the request	
Reader ID	01	Reader ID	
Data Field	62	Data 0 Version	Firmware Version 6V2
	40	Data 1 Status	ST0 Door Relay On (Refer to 2.1.1)
	00	Data 2	Setting Forced Open Alarm
	10	Data 3	Bit Selection (20*xxx#)
XOR	CF	XOR=FF^00^03^01^62^40^00^10 =CF	
SUM	85	SUM=(00+03+01+62+40+00+10+CF) · FF=85(LSB)	

2.8 23H Write Device Real Time Clock

Echo	Example Value	Description
Head	7E	Leading Code
Length	0B	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	23	Writing in the device time
SEC	00	Second
MIN	01	Minute
HR	02	Hour
WEEK	03	Day of a week
DAY	04	Date
MON	05	Month
YEAR	06	Year
XOR	DA	$XOR = FF \oplus 01 \oplus 23 \oplus 00 \oplus 01 \oplus 02 \oplus 03 \oplus 04 \oplus 05 \oplus 06 = DA$
SUM	13	$SUM = (01 + 23 + 00 + 01 + 02 + 03 + 04 + 05 + 06 + DA) \cdot FF = 13(LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.9 24H Read Device Real Time Clock

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	24	Reading out the device time
XOR	DA	$XOR = FF \oplus 01 \oplus 24 = DA$
SUM	FF	$SUM = 01 + 24 + DA = FF$

Echo	Value	Description	
Head	7E	Leading Code	
Length	11	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Function	03	Response to the request	
Reader ID	01	Reader ID	
Data Field	0A	Data 0	Second
	16	Data 1	Minute
	0D	Data 2	Hour
	06	Data 3	Day of a week. Range from 1 to 7 (SUN~SAT)
	09	Data 4	Date
	0C	Data 5	Month
	05	Data 6	Year. Range from 0 to 99 (2000~2099)
	63	Data 7	Firmware Version 6V3
	27	Data 8	Door Number High byte (Used for 701Client huge door mode)
	01	Data 9	Door Number Low Byte
	01	Data 10	Firmware Identify Code (Ver6.6 and later) 00: Is standard firmware
	26	Data 11	Reader Type
XOR	AE	$XOR = FF \oplus 00 \oplus 03 \oplus 01 \oplus 0A \oplus 16 \oplus 0D \oplus 06 \oplus 09 \oplus 0C \oplus 05 \oplus 63 \oplus 27 = AE$	
SUM	89	$SUM = (00 + 03 + 01 + 0A + 16 + 0D + 06 + 09 + 0C + 05 + 63 + 27 + AE) \cdot FF = 89(LSB)$	

2.10 27H Set text on LCD display

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	16	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	27	Stopping waiting for response and Setting the display content of LCD ,display time and beep sounds, each line can be put 16 characters and ASCII is asked (AR-727H only)	
Column	30	LCD Coordinates	LCD coordinates from 00 to 63 Column: 00 — 15 First Row (0x00 — 0x0F) Column: 16 — 31 Second Row (0x10 — 0x1F) Column: 32 — 47 Third Row(0x20 — 0x2F) Column: 48 — 63 Forth Row(0x30 — 0x3F)
Bytes	04	Length of the iuput dat bytes	
Data	54	ASCII code of T	
	6F	ASCII code of O	
	6E	ASCII code of N	
	79	ASCII code of Y	
Delay		High byte of delay time ,unit of 10ms. Available for version 6.5 and later	
		Low byte of delay time ,unit of 10ms.	
Beeps		Beep sounds. (Range from 1 ~ 63) . Available for version 6.5 and later	
XOR	C1	XOR=FF^01^27^30^04^54^6F^6E^79 =C1	
SUM	C7	SUM=(01+27+30+04+54+6F+6E+79+C1) · FF=C7(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

2.11 25H Get the lasted event Log of Device

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicates which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	25	Reading events from the device
XOR	DB	$XOR = FF \oplus 01 \oplus 25 = DB$
SUM	01	$SUM = (01 + 25 + DB) \cdot FF = 01 (LSB)$

2.11.1 Events

Events	Description	Note
01 H	PIN Error	
03 H	Invalid Card	
04 H	Time Zone Error	No entry at this time zone
0B H	Normal Access	
10 H	Egress	Request to exit
11 H	Alarm Event	
1E H	Anti-pass-back Error	

*) For more information take reference to "MessageFunctionCode.PDF" and "Message File structure.pdf".

2.11.2 Event Sources

Value	Description
F0 H	The event is caused by the device
F1 H	The event is caused by the auxiliary reader

2.11.3 Duty Key value

Value	Bit 7~5	Note
00	000 : On Duty	
20	001 : Off Duty	
40	010 : On OVT	Work Overtime On
60	011 : Off OVT	Work Overtime Off
80	100 : Lunch Out	
A0	101 : Lunch In	
C0	110 : Exit	
E0	111 : Return	

(1)No Events

Echo	Value	Description				
Head	7E	Leading Code				
Length	05 (07/0A/0B)	Data Length Indicators which denotes the length from Node to the end including XOR and SUM. AR721W/D will followed IO status				
Node	00	The value 00 is fixed, the message would be sent to PC from the device				
Function	04	Command Acknowledge (ACK)				
Reader ID	01	Reader ID				
		Followed data will be appended after 2009.FEB.09				
Reader type	81	721HB(21H), 72HD(22H), 727HB(23H), 727HD(24H) 81H:721W, 82H:721D, 83H:721Q (For Version 6.3 and later, In old version the reader always 00H)				
Data 0	XX	IO Status0				
Data 1		721Q	721W	721D	721/727H	
		tag bit31~24			IO Status1	
Data 2		tag bit 23~16			parameters	
Data 3		tag bit 15~08			F/w Version	
Data 4		tag bit 07~00				
XOR	FA	XOR=FF^00^04^01 =FA				
SUM	FF	SUM=00+04+01+FA=FF				

1. In AR721H only function code (04H) will be echo.

In AR721W/D the IO status will be followed.

IO Status 0	AR721W	AR721/321/323D	AR721Q	721H/727H
Bit 7	Relay On/Off (1/0)	Low Battery (1/0)		KBD Locked
Bit 6	Card Present (1/0)	Lock working (1/0)		Door Relay On
Bit 5	DO_1 On/Off (1/0)	n/a		Alarm Relay On
Bit 4	DO_0 On/Off (1/0)			ARMED
Bit 3	DI-3 Status	Internal Locked (1/0)		Alarming
Bit 2	DI-2 Status	Force Open Alarm		Door Relay Manual Latched
Bit 1	DI-1 Status	Lock Sensor		PTE pushed
Bit 0	DI-0 Status	Door Sensor		Door opened
IO Status 1				
Bit7				Force open
Bit6				
Bit5				Editing
Parameters				Command 20*

If there have card inserted on AR721W, the Bit6 will be set and Card UID will followed after status0

(2)Invalid Cards

Echo	Value	Description	
Head	7E	Leading Code	
Length	1D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Event	03	Invalid Cards	
Reader ID	01	Reader ID	
Data Field	01	Data 0	Second
	35	Data 1	Minute
	01	Data 2	Hour
	02	Data 3	Day of a week
	05	Data 4	Date
	06	Data 5	Month
	06	Data 6	Year
	01	Data 7	Represent the message source ID
	04	Data 8	Addr Hi
	00	Data 9	Addr Lo
	00	Data 10	On Duty
	00	Data 11	Bit7 : Forced Open Alarm
	18	Data 12	Bit Selection (20*xxx#)
	00	Data 13	Represent the message source comes from WG, bit7 sets 1.
	63	Data 14	Site Hi
	6B	Data 15	Site Lo
	01	Data 16	Door Number (Same as Node)
	14	Data 17	401RO16 Parameter Setting (24*xxx#)
	B8	Data 18	Card Hi
	B4	Data 19	Card Lo
If SOR money deduct off, then this 4 bytes are meaningless	00	Data 20	Deducted amount
	00	Data 21	
	00	Data 22	Balance
	00	Data 23	
XOR	C3	XOR=FF^00^03^01^01^35^01^02^05^06^06^01^04^00^00^00^18^00^63^6B^01^14^B8^B4^00^00^00^00=C3	
SUM	7D	SUM=(00+03+01+01+35+01+02+05+06+06+01+04+00+00+00+18+00+63+6B+01+14+00+00+00+00+C3) · FF=7D(LSB)	

(3)Valid Card

Echo	Value	Description	
Head	7E	Leading Code	
Length	1D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM	
Node	00	The value 00 is fixed, the message would be sent to PC from the device	
Event	0B	valid Cards	
Reader ID	01	Reader ID	
Data Field	36	Data 0	Second
	21	Data 1	Minute
	02	Data 2	Hour
	02	Data 3	Day of a week
	05	Data 4	Date
	06	Data 5	Month
	06	Data 6	Year
	01	Data 7	Represent the message source ID
	00	Data 8	Addr H
	01	Data 9	Addr L
	00	Data 10	On Duty
	00	Data 11	Bit7 : Forced Open Alarm
	18	Data 12	Bit Selection (20*xxx#)
	00	Data 13	Represent the message source comes from WG, bit7 sets 1.
	63	Data 14	Site Hi
	6B	Data 15	Site Lo
	01	Data 16	Door Number (Same as Node)
	14	Data 17	401RO16 Parameter Setting (24*xxx#)
	B8	Data 18	Card Hi
	B4	Data 19	Card Lo
If SOR money deduct off, then this 4 bytes are meaningless	00	Data 20	Deducted amount
	00	Data 21	
	00	Data 22	Balance
	00	Data 23	
XOR	EE	XOR=FF^00^0B^01^36^21^02^02^05^06^06^01^00^01^00^00^18^00^63^6B^01^14^B8^B4^00^00^00^00 =EE	
SUM	CF	SUM=(00+0B+01+36+21+02+02+05+06+06+01+00+01+00+00+18+00+63+6B+01+14+B8+B4+00+00+00+00+EE) · FF=CF	

2.12 37H Delete the lasted transaction

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	37	Clearing events from the device
XOR	C9	$\text{XOR} = \text{FF} \wedge 01 \wedge 37 = \text{C9}$
SUM	01	$\text{SUM} = (01 + 37 + \text{C9}) \cdot \text{FF} = 01 (\text{LSB})$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	$\text{XOR} = \text{FF} \wedge 00 \wedge 04 \wedge 01 = \text{FA}$
SUM	FF	$\text{SUM} = 00 + 04 + 01 + \text{FA} = \text{FF}$

2.13 2DH Empty Device transaction queue

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	2D	Clearing all events from the device
XOR	D3	$XOR = FF \oplus 01 \oplus 2D = D3$
SUM	01	$SUM = (01 + 2D + D3) \cdot FF = 01 (LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.14.1 30H Display bitmap patten onto LCD (AR727H Only)

Echo	Example Value	Description
Head	7E	Leading Code
Length	0D	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command	30	Show bitmap data to AR727H LCD (for f/w version 4.4 and later only)
Bytes	08	8 Bytes (Max. is 10 bytes)
Data 0	11	Byte 1
Data 1	22	Byte 2
Data 2	33	Byte 3
Data 3	44	Byte 4
Data 4	55	Byte 5
Data 5	66	Byte 6
Data 6	77	Byte 7
Data 7	88	Byte 8
XOR	4E	$XOR = FF \oplus 01 \oplus 30 \oplus 08 \oplus 11 \oplus 22 \oplus 33 \oplus 44 \oplus 55 \oplus 66 \oplus 77 \oplus 88 = 4E$
SUM	E8	$SUM = (01 + 30 + 08 + 11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + 4E) \cdot FF = E8(LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.14.2 30H Pass data to TTL serial port of AR721H

Echo	Example Value	Description
Head	7E	Leading Code
Length	0D	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command	30	Pass data to TTL serial output port
Bytes	08	8 Bytes (Max. is 10 bytes)
Data 0	11	Byte 1
Data 1	22	Byte 2
Data 2	33	Byte 3
Data 3	44	Byte 4
Data 4	55	Byte 5
Data 5	66	Byte 6
Data 6	77	Byte 7
Data 7	88	Byte 8
XOR	4E	$XOR = FF \oplus 01 \oplus 30 \oplus 08 \oplus 11 \oplus 22 \oplus 33 \oplus 44 \oplus 55 \oplus 66 \oplus 77 \oplus 88 = 4E$
SUM	E8	$SUM = (01 + 30 + 08 + 11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + 4E) \cdot FF = E8(LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

Ex: Send command 21H to AR401RO16(Node:01H) of AR721H(Node:02H)

Tx: 7E 0C 02 30 07 [01 17 00 80 08 40 04] 10 2D

The TTL serial port will send : 7E 09 01 17 00 80 08 40 04 25 09 to AR401RO and active Relay 15, 3 and inactive relay 14, 2.

2.15 31H Mifare Complex Command

Echo	Example Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	31	Mifare Complex Command which allows to write 16 bytes into assigned KeyA or KeyB of Block if the Device Key is matched
Mifare Command Code	13	Reading data from Mifare cards (Please refer to the protocol of AR-737U1356)
XOR	DC	$XOR = FF \oplus 01 \oplus 31 \oplus 13 = DC$
SUM	21	$SUM = (01 + 31 + 13 + DC) \cdot FF = 21(LSB)$

Echo	Value	Description		
Head	7E	Leading Code		
Length	0D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM		
Node	00	The value 00 is fixed, the message would be sent to PC from the device		
Function	03	Response to the request		
Reader ID	01	Reader ID		
Data Field	69	Data 0	Site Hi	Site Hi Site Lo=0x69AF (27055)
	AF	Data 1	Site Lo	
	F5	Data 2	Card Hi	Card Hi Card Lo=0xF504 (62724)
	04	Data 3	Card Lo	
	01	Data 4	Type of Cards (Please refer to the protocol of AR-737U1356)	
	00	Data 5	UID6	
	7A	Data 6	UID5	
	12	Data 7	UID4	
XOR	A3	XOR=FF^00^03^01^69^AF^F5^04^01^00^7A^12 =A3		
SUM	45	SUM=(00+03+01+69+AF+F5+04+01+00+7A+12+A3) · FF=45(LSB)		

2.16 80H Setting Node ID

Echo	Example Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	80	Setting Node ID
Data	02	New ID
Data	44	ASCII 'D' represent that this command only suitable for versions after AR721H 6.0 and AR727H 7.0
Data	DoorNumH	MSB of Door Group Number, with ranges of 1~254
Data	DoorNumL	LSB of Door Group Number, with ranges of 1~254
Data	Relay Time	High byte of door relay delay time(Base on 10ms)
Data	Relay Time	Low byte of door relay delay time(Base on 10ms)
XOR	7C	$XOR = FF \oplus 01 \oplus 80 \oplus 02 = 7C$
SUM	FF	$SUM = 01 + 80 + 02 + 7C = FF$

Versions after AR721H 6.0 and AR727H 7.0 can set 2 Bytes Door Number, otherwise it can only set Destination ID

When AR721H connect under AR716E, the Door Number build from AR716E's Destination ID as the MSB(Most Significant Bit) and the Door Number as the LSB(Lest Significant Bit).

When AR721H connect to computer, the Door Number builds from its own DoorNumH as MSB and DoorNumL as LSB.

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	02	Reader ID
XOR	F9	$XOR = FF \oplus 00 \oplus 04 \oplus 02 = F9$
SUM	FF	$SUM = 00 + 04 + 02 + F9 = FF$

2.17 81H Resetting Device

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	81	Resetting the device (AR-727H only)
XOR	7F	$XOR = FF \oplus 01 \oplus 81 = 7F$
SUM	01	$SUM = (01 + 81 + 7F) \cdot FF = 01 (LSB)$

Note: This command belongs to unresponsive commands which wouldn't respond either correct or wrong information.

2.17.1 Set factory default and reset reader

Echo	Example Value	Description
Head	7E	Leading Code
Length	07	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	81	Resetting the device (AR-727H only)
Dat0	46	ASCII 'F'
Dat1	41	ASCII 'A'
Dat2	43	ASCII 'C'
XOR		
SUM		

2.18 82H Setting Time & Attendance

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	05	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	82	Setting Time & Attendance (AR-727H only)	
Duty	01	Setting the status "Off Duty"	0~7 refer to different status
			00 On Duty
			01 Off Duty
			02 On OVT
			03 Off OVT
			04 Lunch Out
			05 Lunch In
			06 Exit
			07 Return
XOR	7D	XOR=FF^01^82^01 =7D	
SUM	01	SUM=(01+82+01+7D) · FF=01(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

2.19 83H Setting Card Content

Echo	Value	Description		
Head	7E	Leading Code		
Length	0E	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM		
Destination ID	01	Node ID of destination, the node id is 01 here		
Command Code	83	Setting card content (AR-721H / AR-727HV3 only)		
Addr H	00	User Address – High	Addr H Addr L=0x01(00001) User Address : 00001	
Addr L	01	User Address - Low		
Site H	04	Site Code – High	Site Hi Site Lo = 0x441 (01089) Site Code 01089	
Site L	41	Site Code – Low		
Card H	EA	Card Code – High	Card Hi Card Lo = 0xEA4B (59979) Card Code 59979	
Card L	4B	Card Code – Low		
PIN H	04	PIN – High	PIN H PIN L=0x4D2(1234) PIN 1234	
PIN L	D2	PIN – Low		
Mode	02	Card or PIN	Access Mode(Bit 1~0)	
			0	Invalid
			1	Read Only
			2	Card or PIN
			3	Card + PIN
		Bit7: Set to enable Anti passback		
Zone	01	Selecting the 1st Time Zone (11 Time Zones for selection)		
Group		Bit mapping of Door Group. (B0:WG1, B1:WG2, B3:WG3, B4:WG4) ※ for AR721E only.		
XOR	xx	XOR=FF^01^83^00^01^04^41^EA^4B^04^D2^02^01 =4D		
SUM	xx	SUM=(01+83+00+01+04+41+EA+4B+04+D2+02+01+4D) · FF=25(LSB)		

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

2.20 87H Reading Card Content

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	07	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	87	Reading the card content	
Addr H	00	User Address – High	Addr H Addr L=0x02(00002) User Address : 00002
Addr L	02	User Address - Low	
Nums	01	Number of cards	
XOR	7A	XOR=FF^01^87^00^02^01 =7A	
SUM	05	SUM=(01+87+00+02+01+7A) · FF=05(LSB)	

Echo	Value	Description		
Head	7E	Leading Code		
Length	0D	Data Length Indicator which denotes the length from Node to the end including XOR and SUM		
Node	00	The value 00 is fixed, the message would be sent to PC from the device		
Function	03	Response to the request		
Reader ID	01	Reader ID		
Data Field	04	Data 0	Site Code – High	Site Hi Site Lo = 0x441 (01089)
	41	Data 1	Site Code – Low	Site Code : 01089
	EA	Data 2	Card Code – High	Card Hi Card Lo = 0xEA4B (59979)
	4B	Data 3	Card Code – Low	Card Code : 59979
	04	Data 4	PIN – High	PIN H PIN L=0x4D2(1234)
	D2	Data 5	PIN - Low	PIN : 1234
	02	Data 6	Access Mode	Card or PIN
	0B	Data 7	Time Zone	11st Time Zone
		Data8	High Tag ID (bit41~38)	For Sony Tag Only
		Data9	High Tag ID (bit37~32)	For Sony Tag Only
XOR	C6	XOR=FF^00^03^01^04^41^EA^4B^04^D2^02^0B =C6		
SUM	27	SUM=(00+03+01+04+41+EA+4B+04+D2+02+0B+C6) · FF=27(LSB)		

2.21 84H Stopping Waiting for Response

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	84	Stopping waiting for response. When the controller receives events from the device (reader), the device will wait for the response- ACK or NACK from the controller, otherwise the device will show error message in 2.5 sec. Note: This command is unresponsive command which wouldn't respond either correct or wrong information.
XOR	7A	$XOR = FF \oplus 01 \oplus 84 = 7A$
SUM	FF	$SUM = 01 + 84 + 7A = FF$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.22 85H Clearing All Card Content

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	85	Clearing all card content, this requests 10 sec. to process (AR-721H / 727HV1 / 721QFM Only)
Option	0x01	Set Bit0 to clear all normal tag ID Set Bit1 to clear all black tag ID *) Only for 721Q(Ver6.6 and later)/323D(Ver6.8 and later)
XOR	7B	$XOR = FF \oplus 01 \oplus 85 = 7B$
SUM	01	$SUM = (01 + 85 + 7B) \cdot FF = 01(LSB)$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.23 86H Resetting Anti-pass-back

Echo	Example Value	Description
Head	7E	Leading Code
Length	04	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	86	Restoring the initial setup of anti-pass-back (AR-721H / 727HV3 only)
XOR	78	$XOR = FF \oplus 01 \oplus 86 = 78$
SUM	FF	$SUM = 01 + 86 + 78 = FF$

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.24 2AH Setting Time Zone

Echo	Example Value	Description			
Head	7E	Leading Code			
Length	24	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM			
Destination ID	01	Node ID of destination, the node id is 01 here			
Command Code	2A	Set Time Zone (AR-721H / 727HV3 only)			
IDX	01	The first time zone index of the download data		The 721H supports 11 timezones. (01h~0Bh) Value=00h :Duty Zone (please refer to 2.24.1)	
Sets	01	How many time zone are followed			
Data 0	0C	Link to the next		Bit7 : Available on holidays Bit6 ~ Bit0 : The next time zone link	
Data 1	00	Time zone level, users are only allowed to pass when their card level > time zone level			
Data 2	01	Sun.	Begin Time (08:00) 0x1E0	32bytes within the data, the first two bytes are Link and Level Data 0 : Link Data 1 : Level Data 2 Data 3 : Beginning time on Sunday 08:00 refers to 8 a.m., Namely 480 minutes (0x1E0) Data 4 Data 5 : End time on Sunday Data 6 Data 9 : Monday Data 10 Data 13 : Tuesday Data 14 Data 17 : Wednesday Data 18 Data 21 : Thursdays Data 22 Data 25 : Friday Data 26 Data 29 : Saturday Data 30 : Reserved Data 31 : Reserved	
Data 3	E0		End Time (15:00) 0x384		
Data 4	03		Beginning Time (08:00) 0x1E0		
Data 5	84		End Time (15:00) 0x384		
Data 6	01	Mon.	Beginning Time (08:00) 0x1E0		
Data 7	E0		End Time (15:00) 0x384		
Data 8	03		Beginning Time (08:00) 0x1E0		
Data 9	84		End Time (15:00) 0x384		
Data 10	01	Tue.	Beginning Time (08:00) 0x1E0		
Data 11	E0		End Time (15:00) 0x384		
Data 12	03		Beginning Time (08:00) 0x1E0		
Data 13	84		End Time (15:00) 0x384		
Data 14	01	Wed.	Beginning Time (08:00) 0x1E0		
Data 15	E0		End Time		
Data 16	03		End Time		

Data 17	84		(15:00) 0x384	
Data 18	01	Thu.	Beginning	
Data 19	E0		Time (08:00) 0x1E0	
Data 20	03		End Time	
Data 21	84		(15:00) 0x384	

[illegible]

2.24.1 Auto-shift Setup

When the value of time zone is 00, it refers to auto-shift		
Beginning Time	End Time	Description
00:00	00:00	Always On Duty
23:59	23:59	Always Off Duty
00:01	00:01	Duty status depends on the last setting and will be fixed on it
Example		
08:00	18:00	(Overtime Setup) Before 08:00→ On Duty , Before 18:00→ Off Duty , After 18:00 Off OVT
08:00	23:59	(Non-overtime Setup) Before 08:00→ On duty , Before 23:59→ Off Duty

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

2.25 2CH Setting Holidays

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	12	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination	01	Node ID of destination, the node id is 01 here	
Code	2C	Setting Holidays (AR-721H / 727HV3 only)	
IDX	00	First set of holidays	120 days available to setup for holiday, sorting before download and put the small date first then follow zero to fill all filed.
Sets	06	Number of holidays, 06 refers to 6-day holiday	
Data 0	0C	December	Set 00
Data 1	01	1	Dec. 1
Data 2	0C	December	Set 01
Data 3	02	2	Dec. 2
Data 4	0C	December	Set 02
Data 5	03	3	Dec. 3
Data 6	0C	December	Set 03
Data 7	04	4	Dec. 4
Data 8	0C	December	Set 04
Data 9	05	5	Dec. 5
Data 10	0C	December	Set 05
Data 11	06	6	Dec. 6
XOR	D3	XOR=FF^01^2C^00^06^0C^01^0C^02^0C^03^0C^04^0C^05^0C^06 =D3	
SUM	63	SUM=(01+2C+00+06+0C+01+0C+02+0C+03+0C+04+0C+05+0C+06 +D3) · FF=63(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

2.26 88H Setting Extend Parameters for AR721H only

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	07	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	88	Setting Extend Parameters (AR-721H / 727HV3 only) EEPROM Address 23(17H)	
Data 0	00	Set to 1 Mask	Set bit 1 to 1 will set Extend Parameter bit1
Data 1	00	Set to 0 Mask	Set bit 1 to 1 will clear Extend Parameter bit1
XOR	D3	XOR=FF^01^2C^00^06^0C^01^0C^02^0C^03^0C^04^0C^05^0C^06 =D3	
SUM	63	SUM=(01+2C+00+06+0C+01+0C+02+0C+03+0C+04+0C+05+0C+06 +D3) · FF=63(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledged (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

Extend Parameter: Version 6V9(727H) / 6V6(721H) and later

Bit7: Reserved

bit6: Reserved

Bit5: Reserved

Bit4: Need password always when access on this device

Bit3: Door Relay Latch state On/Off(1:0)

Bit2: Latch Door relay status(Keep the last status depend on bit3)

Bit1: Set to 1 disable egress function while bit0 set.

Bit0: Set to 1 will disable open door function after tag access passed, but event log will generate normally.

And Just show [Tag Locked] message (3 beep sounds)

2.27 90H Lock Indicator (LCD display / LED / Keyboard) for AR727H only

Echo	Example Value	Description
Head	7E	Leading Code
Length	16	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM
Destination ID	01	Node ID of destination, the node id is 01 here
Command Code	90	Setting the display content of LCD, the panel would light for 6.5 sec.. 4 rows on the panel, each line can be put 16 characters and ASCII is asked (AR-727H only)
Selection	30	Bit0: 1/0 Lock/Free LCD display Bit3: 1/0 Lock/Free LEDs Bit4: 1/0 Lock/Free Keyboard
Lock Time	00	High byte of lock time. (Base on 10ms.)
Lock Time	80	Low byte of lock time.
XOR		
SUM		

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	$XOR = FF \oplus 00 \oplus 04 \oplus 01 = FA$
SUM	FF	$SUM = 00 + 04 + 01 + FA = FF$

*) After keyboard locked, all pressed key will store in buffer, and send to host during polling status command(18H) with status code 06H..

Ex: 7E LED DID 09 SID [06] [current Key] [Key bytes] [Key...] XOR SUM

Current Key: 0x80 mean keyboard free, 0x88 mean key still pressed

Key bytes: How many bytes of key data followed.

2.28 89H Insert Tag Into Device by UID without address assigned

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	0C	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	89	Insert Tag Into Device by UID without address assigned (Only 721Q Ver6.6 and later supported)	
Site H	04	Site Code – High	Site Hi Site Lo = 0x441 (01089)
Site L	41	Site Code – Low	
Card H	EA	Card Code – High	Card Hi Card Lo = 0xEA4B (59979)
Card L	4B	Card Code – Low	
PIN H	04	PIN – High	PIN H PIN L=0x4D2(1234)
PIN L	D2	PIN - Low	
Mode	02	Card or PIN	Access Mode
			00 Invalid
			01 Read Only
			02 Card or PIN
			03 Card + PIN
Zone	01	Selecting the 1st Time Zone (11 Time Zones for selection)	
XOR	4D	XOR=FF^01^83^00^01^04^41^EA^4B^04^D2^02^01 =4D	
SUM	25	SUM=(01+83+00+01+04+41+EA+4B+04+D2+02+01+4D) · FF=25(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

*) Take reference to command 83H please

2.29 8AH Delete Tag from Device by UID without address assigned

Echo	Example Value	Description	
Head	7E	Leading Code	
Length	08	Data Length Indicator which denotes the length from Destination ID to the end including XOR and SUM	
Destination ID	01	Node ID of destination, the node id is 01 here	
Command Code	8A	Delete Tag from Device by UID without address assigned (Only 721Q Ver6.6 and later supported)	
Site H	04	Site Code – High	Site Hi Site Lo = 0x441 (01089) Site Code 01089
Site L	41	Site Code – Low	
Card H	EA	Card Code – High	Card Hi Card Lo = 0xEA4B (59979) Card Code 59979
Card L	4B	Card Code – Low	
XOR	4D	XOR=FF^01^83^00^01^04^41^EA^4B^04^D2^02^01 =4D	
SUM	25	SUM=(01+83+00+01+04+41+EA+4B+04+D2+02+01+4D) · FF=25(LSB)	

Echo	Value	Description
Head	7E	Leading Code
Length	05	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	04	Command Acknowledge (ACK)
Reader ID	01	Reader ID
XOR	FA	XOR=FF^00^04^01 =FA
SUM	FF	SUM=00+04+01+FA=FF

3 Data Structure

3.1 Data Structure of Cards :

```

Struct    DS_CARD    {
    Unsigned int    site ;           //Part of Tag UID ( bit 31~16 )
    Unsigned int    map ;           //Part of Tag UID ( bit 15~0 )
    Unsigned int    pin ;           //PIN
    Unsigned char    zone:6 ;       //Time zone ( Range 0~11 )
                                     // 0=Auto-shift

    Unsigned char    mode2 ;        //Access Mode
                                     // 0:Invalid      1:CardOnly
                                     // 2:Card & PIN    3:Card or PIN

    Unsigned char    rev ;          //Reserved
#ifdef    SONY_TAG
    Unsigned int    hUID;           // higher bits of 42 bits TAG
#endif
};

```

Ultra Hi 4 bits for Sony Tag	
Ultra Hi 8 bits for Sony Tag	
Site Hi 8 bits	
Site Lo 8 bits	
Card Hi 8 bits	
Card Lo 8 bits	
PIN H 8 bits	
PIN L 8 bits	
Time zone 6 bits	Mode 2 bits
Reserved	

3.2 Structure of Time Zone :

```

Struct DS_ZONE {
    Unsigned char link ;           //Bit 7 refers to free pass on holidays

    Unsigned char level ;         //While card level < ZONE level → Pass Denied

    Unsigned int16 Time[7][2]     //Minute as a unit to set
    Unsigned char rev[2]
};

```

Data 0 : Link				
Data 1 : Level				
	Beginning Time		End Time	
	High	Low	High	Low
Sunday	Data 2	Data 3	Data 4	Data 5
Monday	Data 6	Data 7	Data 8	Data 9
Tuesday	Data 10	Data 11	Data 12	Data 13
Wednesday	Data 14	Data 15	Data 16	Data 17
Thursday	Data 18	Data 19	Data 20	Data 21
Friday	Data 22	Data 23	Data 24	Data 25
Saturday	Data 26	Data 27	Data 28	Data 29
Data 30 : Reserved				
Data 31 : Reserved				

11 time zones available to setup, Time Zone 0 refers to Auto-shift

3.3 Data Structure of Holidays :

```
Struct DS_HOLI {
    Unsigned char    month ;
    Unsigned char    data ;
};
```

Set 00		Set 01			Set xx	
Month	Date	Month	Date		Month	Date
Data 0	Data 1	Data 2	Data 3		Data xx	Data xx

120 days available to setup for holiday

3.4 Data Structure of Anti-pass-back:

//Bit mapping to each user. Bit0 of byte0 is user 0 , bit7 of byte1 is user 15,etc.

//If set Enable[0] to 0xFF , the user from 0~7 need anti-pass-back check.

//If set Initial[0] to 0xFF , the user from 0~7 can access reader in free anti-pass-back

// state one times.

Struct DS_ANTIPASS {

Unsigned char Enable[1024 /8] ; //Start from EEPROM address 8448(Dec)

Unsigned char Initial[1024 /8] ; // Start from EEPROM address 8576(Dec)

};

Enable

1024 bits (1024 users)																							
Byte1 (8 users)								Byte2								Byte3							
User Address 7~0								User Address 15~8								User Address 31~16							
0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	1	0	1	1	1	0	0

1 : Enable Anti-pass-back

0 : Disable Anti-pass-back

Initial

1024 bits (1024 users)																							
Byte1 (8 users)								Byte1 (8 users)								Byte1 (8 users)							
User Address 7~0								User Address 7~0								User Address 7~0							
0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	1	0	1	1	1	0	0

1 : Initial Enabling

0 : Non-initial Enabling

Current state

1024 bits (1024 users)																							
Byte1 (8 users)								Byte1 (8 users)								Byte1 (8 users)							
User Address 7~0								User Address 7~0								User Address 7~0							
0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	1	0	1	1	1	0	0

1 : Entry Door

0 : Exit Door

3.5 Memory Layout of AR-721H / AR-727H :

The capacity of EEPROM of AR-721H / AR-727H can be up to 256Kbits. The first 30 bytes are system parameters as

tabled below

Address	Description	Function	Bytes
00h	System Flag	Reserved	6
06h	Function Option 0	Command : 20 *	1
07h	Function Option 1	Command : 28 *	1
08h	Node ID	Command : 00 *	1
09h	Reader Type		1
0Ah	Arming Pulse Output Time	For 327/727/747 Only	1
0Bh	Controller Operation Mode	Must be 4, 6, or 8	1
0C~0Dh	Duress Password		2
0E~11h	Master Password		4
12~13h	Door Relay Time	10ms	2
14~15h	Alarm Relay Time	10ms	2
16h	Function Option 2	Command : 24 * (721H)	1
17h	Extend Parameter	Command 88H	1
18~19h	Arming Delay Time	10ms	2
1A~1Bh	Alarm Delay Time	10ms	2
1C~1Dh	Door Close Delay Time	10ms	2
1E~1Fh	Arming Password		2
26 ~ 27h	Start time of Open Zone Set 1	Base on Minute: 01:20 = 0050h	
28 ~ 29h	Stop time of Open Zone Set 1	"	
2A ~ 2Bh	Start time of Open Zone Set 2	"	
2C ~ 2Dh	Stop time of Open Zone Set 2	"	
2Eh	Weekly table bit of Open Zone1	Available on weekday table Bit1 for Sun, bit7 for Sat	
2Fh	Weekly table bit of Open Zone2		
30~31H	Manager User Start & End	30:Start, 31:End	2
32h			1
33H	Door Number Hi		
34H	Door Number		
35~36H	Tag Interval	Base on 10ms	
3BH	Function Option 3	Command : 34 *	
66~B5H	Auto Alarm schedule	(Hr+Min+Weekday+Sec) * 20	

Address	Description	Function	Bytes
F0~F3h	Customer Define Area	User define sn# or security code	4
FA~FBh	Tag Interval for same ID (Base 10ms)	Default is 100(1000ms), Change the value for different Tag interval	2
FC~FFh	Customer Define Area	User define sn# or security code	4

	Function Option 0 (20*) EEPROM Addr:06h	Function Option 1 (28*) EEPROM Addr:07h
B7	Enabling Anti-pass-back	Forced Open Alarm
B6	0 : Exit Door 1 : Entry Door	Open different door for WG reader and main reader ! (721H Only)
B5	1 : Master Reader	0 (Fixed)
B4	1 : Enabling RTS button (Request to Exit)	0 (Fixed)
B3	0 (Fixed)	0 : Month / Date 1 : Date / Month (727H Only)
B2	1/0: En/Disable Auto-open Time zone(63) Local & AR716E(Time Zone 63)	0 : English 1 : Chinese (727H Only)
B1	Auto Locking when Door Closed	Display ID format:
B0	Time & Attendance 0 : Enabled 1 : Disabled	00 : None 01 : WG 10 : ABA 11 : Hexadecimal (727H Only)

	Function Option 2 (24*) EEPROM Addr:16h	
B7	Enable Bell Function(#)	
B6	Enable Close Door /Egress Will Stop Alarm	
B5	Global Free Cards	
B4/3	Terminal port format (B4:B3) <i>Always 0 for 721QFM</i>	00: Lift Control 01: LED display 10: Printer 11: Duress output
B2	Set 1 to freeze Controller (Stop functions)	
B1	Set 1 to Enable Lift Output and disable Duress output	(727H)
B0	Enable Door relay will active at open time zone, don't wait card flash	

	Function Option 3 (34※) EEPROM Addr:3Bh	
B7	Trigger Security Relay Board by Duress output Pin.	
B6	Trigger Security Relay Board by Door relay output Pin.	
B5	Continue Beeps while alarm state.	
B4		
B3	Disable PTE (Egress) Beep.	
B2	Disable Beep.	
B1	Enable Error Alarm. Any Card Error will on Alarm relay	
B0	Enable Door Close then active RF	

User Data of 1024 user version:

Address	Function		bytes
0100h~10FFh	UID of User 0 to 1023	Each UID has 32bits	4096
1100h~1800h	PIN of User 0 to 1023	Each PIN has 16bits	2048
1900h~20FFh	Time Zone and Access Mode		384
2100h~217Fh	Enable flag of Anti-pass-back		128
2180h~21FFh	Initial flag of Anti-pass-back		128
2280h~22FFh	Current State flag of Anti-pass-back		128
2280h~23FFh	Time ZONE DB from 0 to 0Bh		384
2400h~24FFh	Holiday table		240
2500h ~ 34FFh	Lift stop information 0x2500 bit0:User0, bit7:User7 0x2501 bit0:User8, bit7:User15 ...	* For 1024 users mode only	
3500h ~ 7FFFh	Message information	727/721H 1024 Users Mode	
3500h ~ F9FFh	Message information	747/575H	
FA00h ~ FBFFh	Idle Image bitmap Buffer	727/747H (Reserved)	

PS. 2048 user version EEPROM layout

Address	Function		bytes
0100h~24FFh		Same as 1024 user version	
2500h~34FFh	UID of User 1024 to 2047	Each UID has 32bits	4096
3500h~3EFFh	PIN of User 1024 to 2047	Each PIN has 16bits	2048
3D00h~44FFh	Time Zone and Access Mode		384
4500h~457Fh	Enable flag of Anti-pass-back		128
4580h~45FFh	Initial flag of Anti-pass-back		128
4600h~467Fh	Current State flag of Anti-pass-back		128
4700h ~ 7FFFh	Message information (912 Records)	727/721H 2048 Version	

PS. 3072 user version EEPROM layout

Address	Function		bytes
0000h~00FFh	System Parameters		256
0100h~30FFh	UID of User 0 to 3071	Each UID has 32bits	12288
3100h~48FFh	PIN of User 0 to 5119	Each PIN has 16bits	6144
4900h~60FFh	Access Mode and Zone		6144
6100h~627Fh	Enable flag of Anti-pass-back		384
6280h~63FFh	Initial flag of Anti-pass-back		384
6400h~657Fh	Current State flag of Anti-pass-back		384
6580h~66FFh	Time ZONE DB from 0 to 63		384
6700h~67F0h	Holiday table		240
	reserved		16
6800h ~ 97FFh	Lift stop information 6800h bit0: Floor0, bit7: Floor7 6801h bit0: Floor8, bit7:Floor15 ...	* For 3072 users mode only	12288
9800h ~ F9FFh	Message information (1568)		25088
FA00h ~ FBFFh	Idle Image bitmap Buffer	727/747H (Reserved)	

Update log:

2006.DEC.30:

1. Update command 18H response code: Page 9.
 - 1.1 Add ultra-light support and fix followed message data field

2007.JAN.04:

1. Update command 18H response code: Page 12.
 - 1.1 Add ultra-light echo for swipe card event.
 - 1.2 Fix append message field error
 - 1.3 Add disable SOR function command [7E 08 01 31 0E F2 00 00 33 65]

2007 JUN 29

1. Update command 88H for latch functions

2008 JUL 13

1. Add command 89/8AH for 721Q

2009.DEC.01

1. Upgrade to 725HB for 3000 Users.